

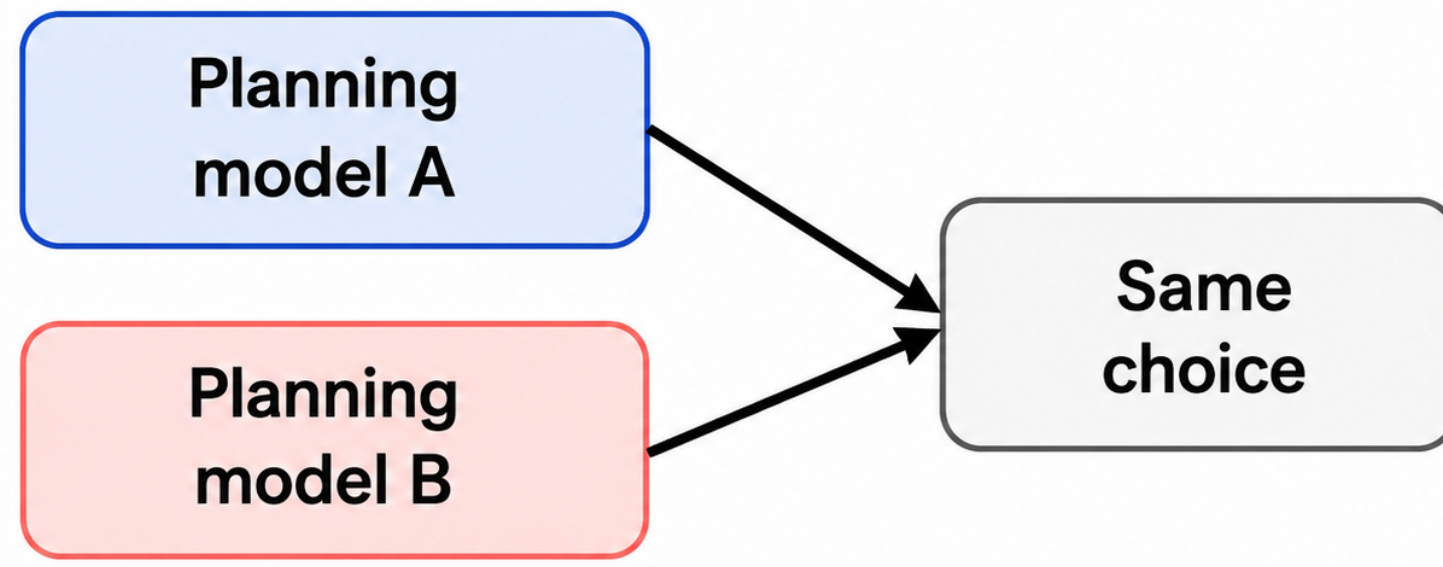
Joint Modeling of Choices and Response Times in Multi-stage Decisions via Likelihood Approximation

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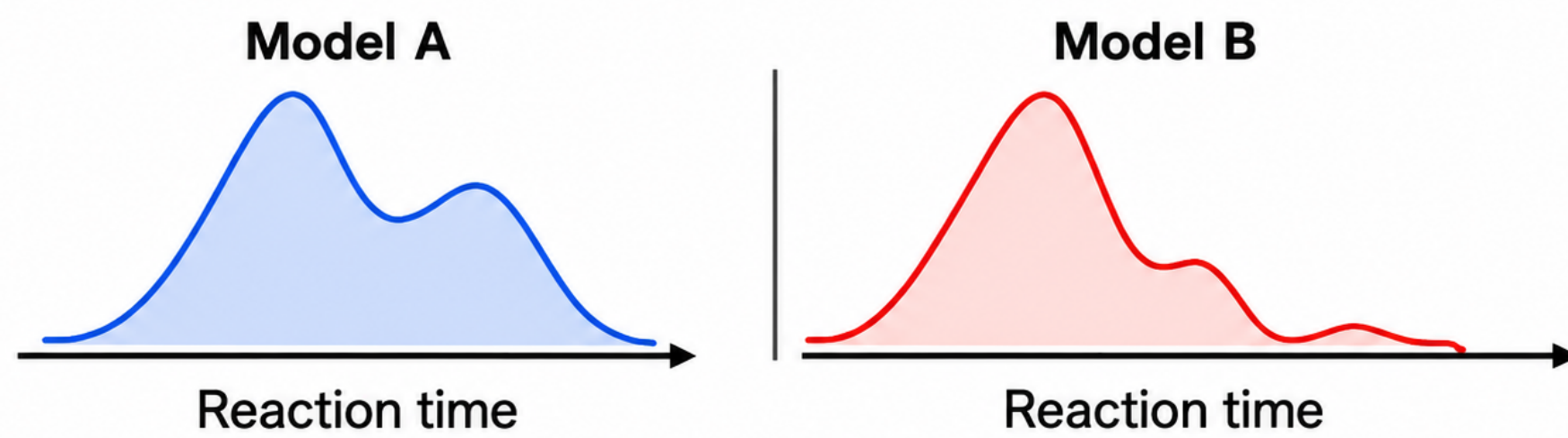
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Motivation

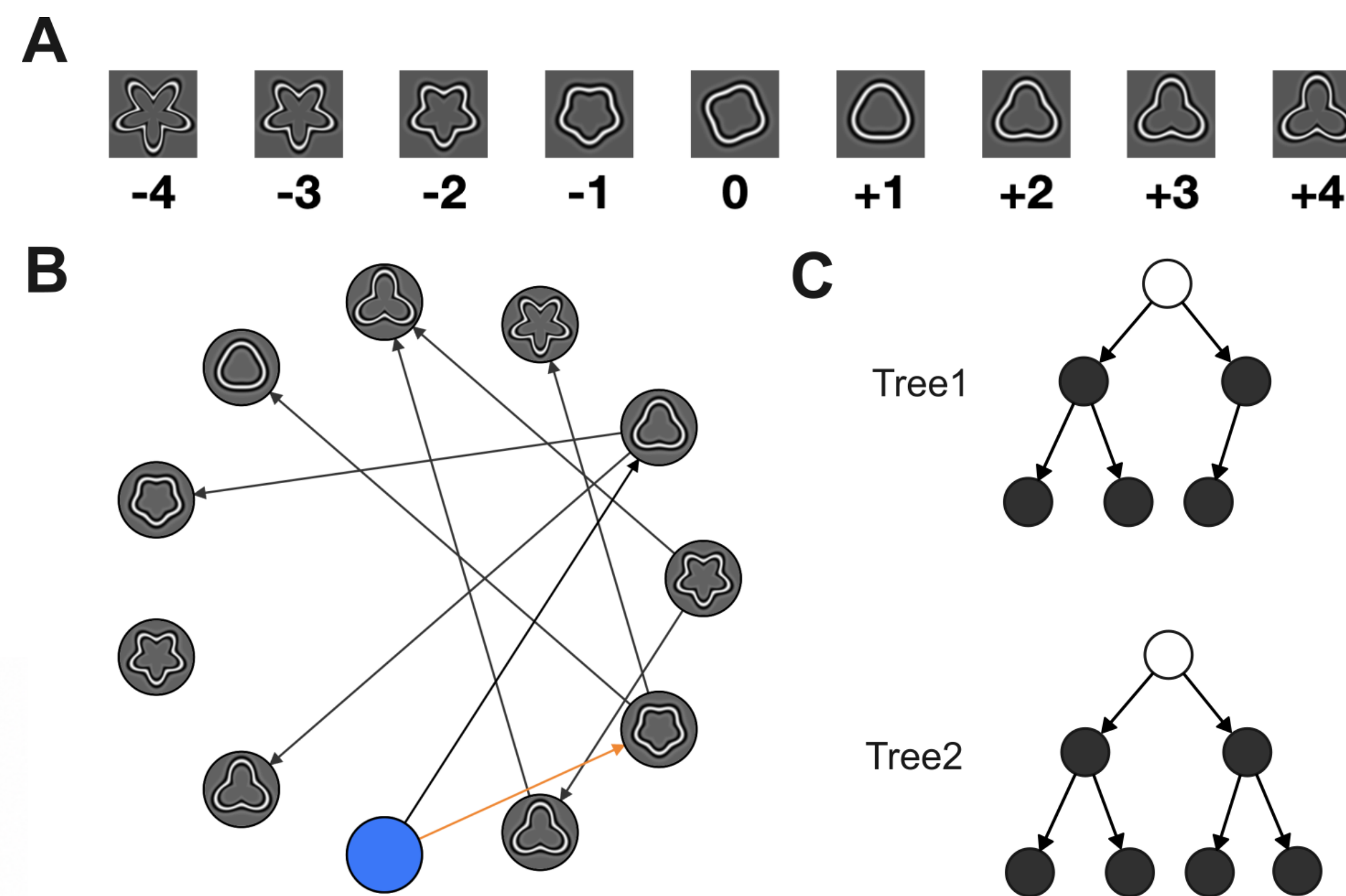
Difference planning algorithms can lead to the same choices, but rely on very different computations



Modeling response time (RT) provide an additional constraints on the forms of computations.



Decision Tree Navigation Task



Computational Models

Forward greedy Model (FG)

$$E_L^{t+1} = E_L^t + d_1 \cdot R_L + \epsilon_L,$$

$$E_R^{t+1} = E_R^t + d_1 \cdot R_R + \epsilon_R,$$

Two stage model (TS)

$$E_{L,L}^{t+1} = E_{L,L}^t + (d_1 R_L + \epsilon_L) + (d_1 R_{LL} + \epsilon_{LL}),$$

$$E_{L,R}^{t+1} = E_{L,R}^t + (d_1 R_L + \epsilon_L) + (d_1 R_{LR} + \epsilon_{LR}),$$

$$E_{R,L}^{t+1} = E_{R,L}^t + (d_1 R_R + \epsilon_R) + (d_1 R_{RL} + \epsilon_{RL}),$$

$$E_{R,R}^{t+1} = E_{R,R}^t + (d_1 R_R + \epsilon_R) + (d_1 R_{RR} + \epsilon_{RR}).$$

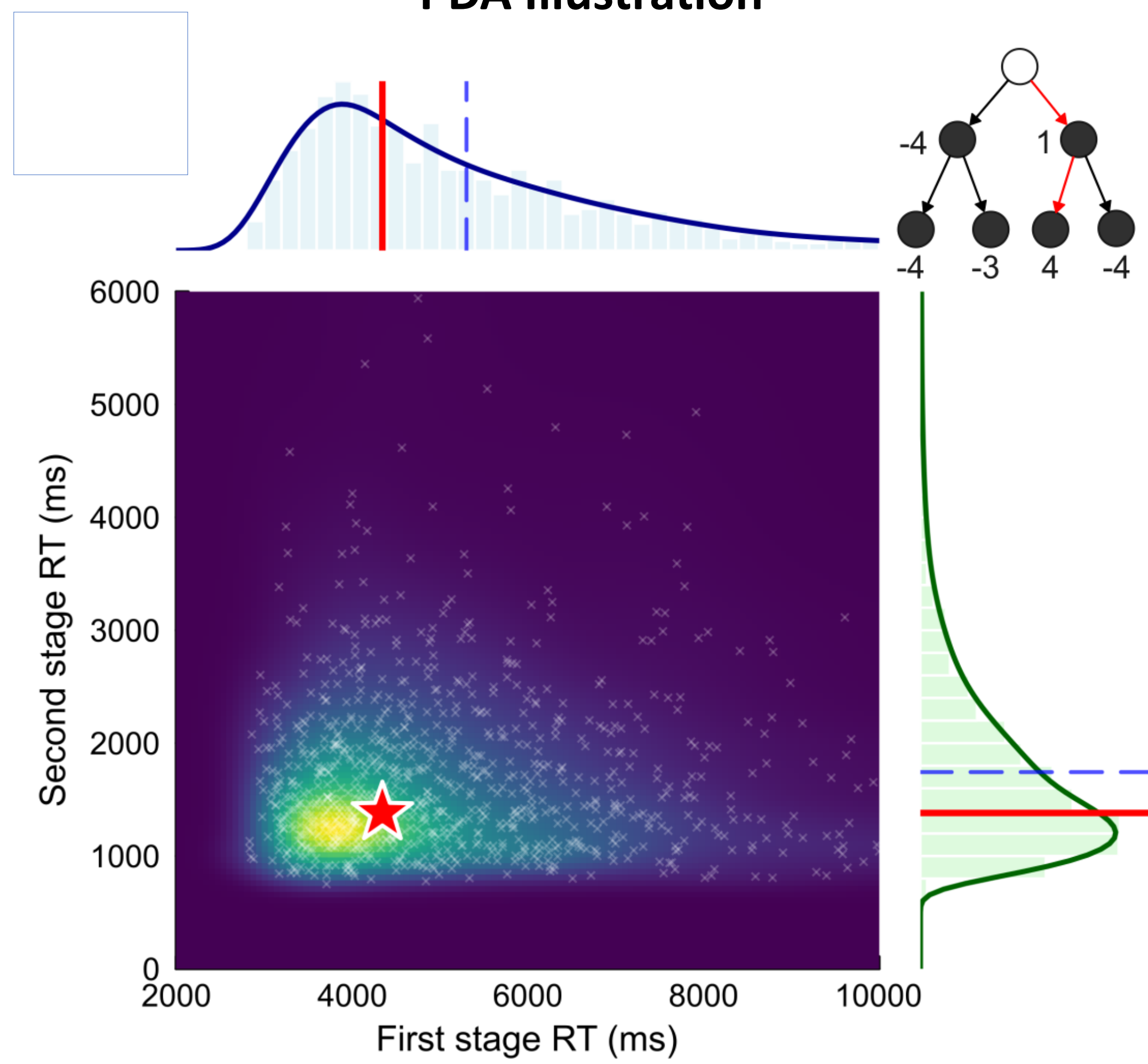
Parameters to be fitted:

d_1 and d_2 : first stage and second stage drift rate
 θ_1 and θ_2 : first stage and second stage threshold
 T_1 and T_2 : first stage and second stage nonddecision time

Full model description can be found in Solway & Botvinick (2015)

Model fitting

PDA Illustration



Probability Density Approximation (PDA)

$$p(c_1, c_2, t_1, t_2 | \theta, s) = p(c_1, c_2 | \theta, s) p(t_1, t_2 | c_1, c_2, \theta, s)$$

$$p(c_1, c_2 | \theta, s) = \frac{n(c_1, c_2)}{J} \quad \hat{f}_H(x) = \frac{1}{n} \sum_{i=1}^n K_H(x - x_i)$$

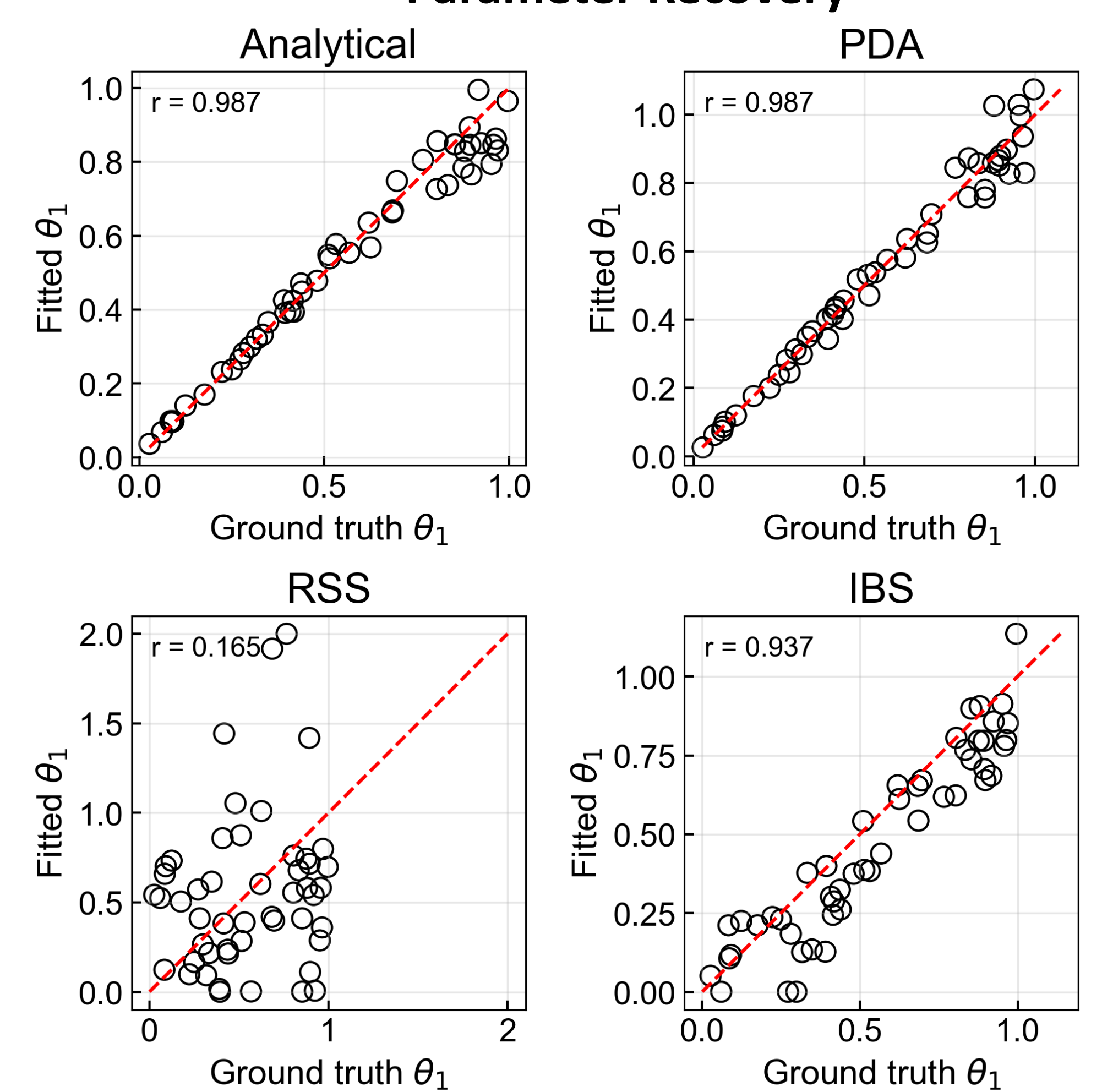
Inverse Binomial Sampling (IBS)

$$\mathcal{L}_\epsilon(\theta) = \sum_{i=1}^N \log \Pr(|\tilde{r}_i - r_i| \leq \epsilon | \theta, s_i)$$

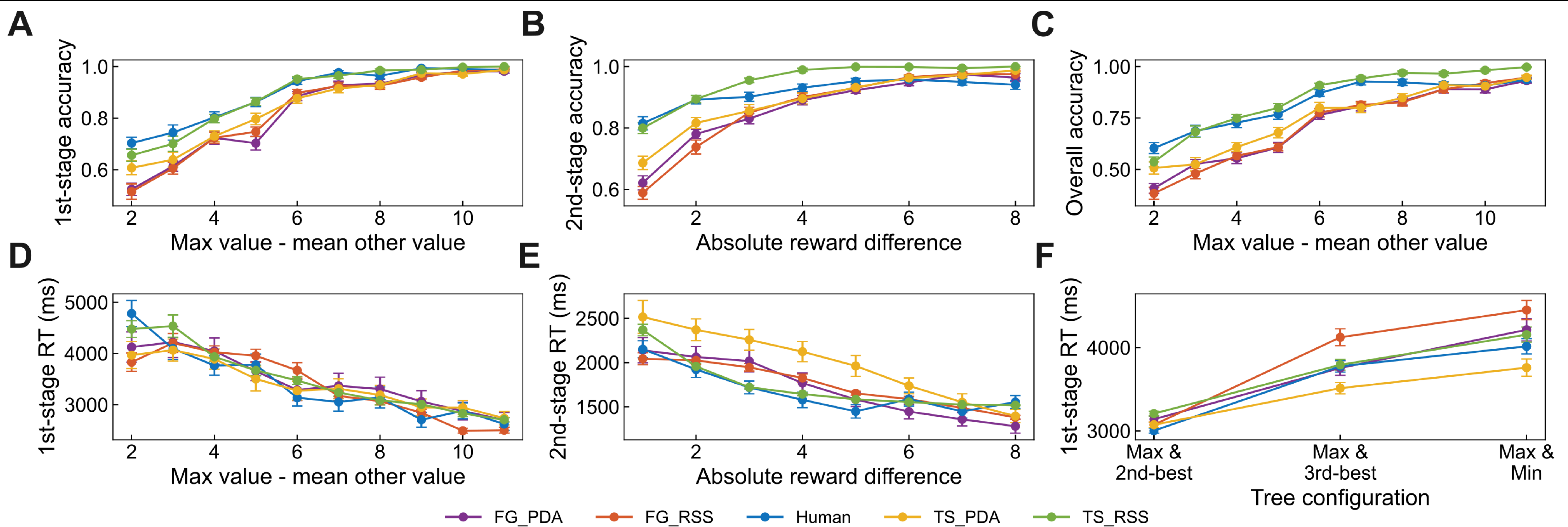
Residual Sum of Squares (RSS)

$$RSS(\theta) = \sum_{j=1}^n (y_j^{\text{data}} - y_j^{\text{sim}}(\theta))^2,$$

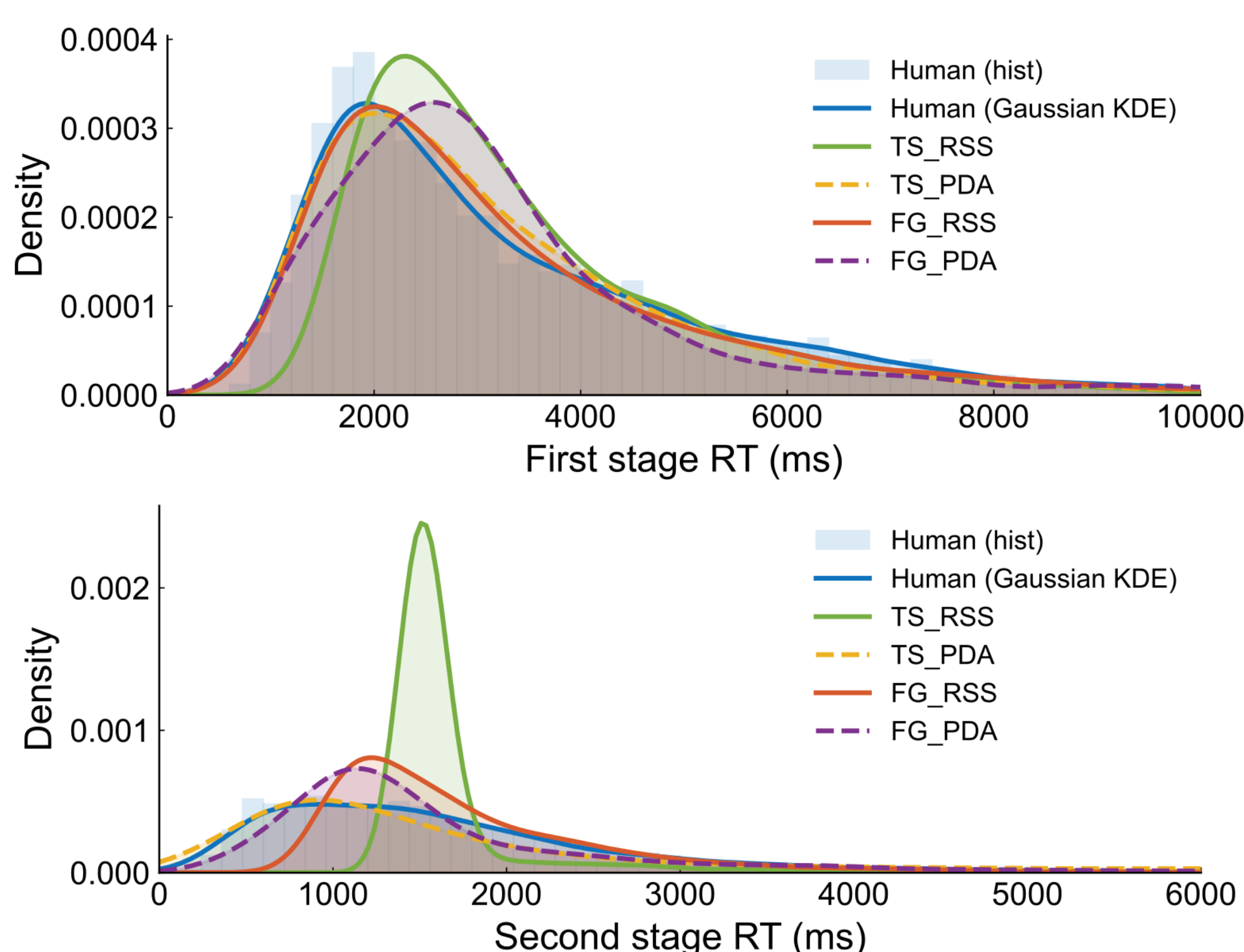
Parameter Recovery



Behavioral Fitting Results



Model Misspecification from Posterior Check



Inconsistent Model Selection (Using Bayesian Information Criteria)

1. RSS method: TS: BIC = -289.11; FG: BIC = -257.96
2. PDA method: TS: BIC = 3556.85; FG: BIC = 3523.47
3. IBS method: TS: BIC = 446.68; FG: BIC = 432.79

Take Home Messages

- ✓ Choice alone is **insufficient** to distinguish planning algorithms.
- ✓ The **full RT distribution** contains critical computational information.
- ✓ **PDA** enables likelihood-based inference for simulation models without analytical likelihoods.
- ✓ Better fitting leads to **different scientific conclusions** about human planning.

References

van Opheusden, B., Acerbi, L., & Ma, W. J. (2020). Unbiased and efficient log-likelihood estimation with inverse binomial sampling. *PLoS computational biology*, 16(12), e1008483.
Solway, A., & Botvinick, M. M. (2015). Evidence integration in model-based tree search. *Proceedings of the National Academy of Sciences*, 112(37), 11708-11713.